How much can each hardpoint carry?

N4 is the strongest normal-holding candidate in the screened sections. For a normal departure, **N3 is the strongest simple candidate to test next**. N5 has slightly lower nominal pull demand, but its contact sensitivity overlaps N3 and its printed hook arm has a lower structural screen.

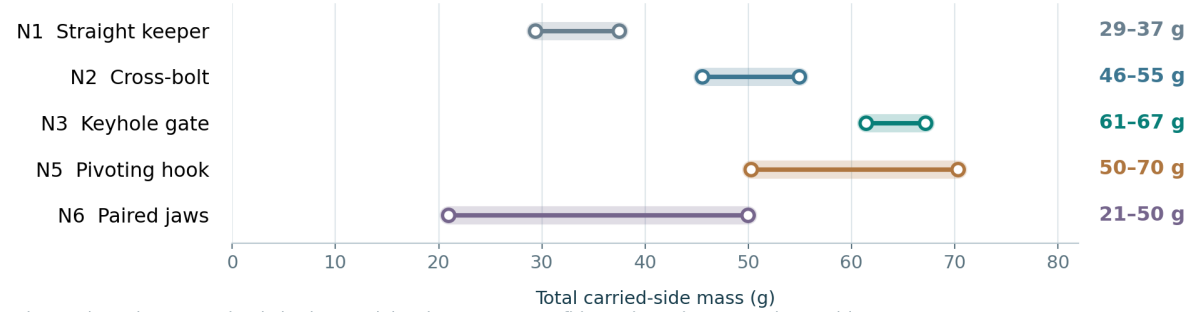
The important practical result is the gap between **holding a load** and **withdrawing the loaded retainer**. The tiny common solenoid can become the limiting component well before the selected static section limit.

Design	Local section screen, N	1g mass equivalent, kg	Release example, total mass
N1 Straight keeper	1.08-8.03	0.11-0.82	29-37 g
N2 Cross-bolt	1.68-21.20	0.17-2.16	46-55 g
N3 Keyhole gate	8.70-14.26	0.89-1.45	61-67 g
N4 T-rail	32.71-65.41	3.34-6.67	Separate sideward extraction
N5 Pivoting hook	4.58	0.47	50-70 g
N6 Paired jaws	6.26-12.51	0.64-1.28	21-50 g

Table assumptions: 10 MPa normal design-stress screen, 6 MPa shear, 10 MPa bearing; release uses μ=0.20 and a hypothetical 0.35 N force allocation at 1g. Ranges span selected contact/load-sharing cases, not confidence intervals or proven bounds. N1/N2 include an assumed 0.5-3 mm roof load-spread width. No mounting or whole-joint capacity is established.

Loaded release favors N3 and N5 in these models

Illustrative friction budget 0.35 N · μ = 0.20 · normal load factor n = 1



Ranges show selected contact / load-sharing models. They are not confidence bounds or tested capacities.
N4 is omitted: its rail needs a separate sideward driving force and more than 17.5 mm of travel.
Mass includes payload, adapter and payload-side hardware. At n = 3, divide the displayed masses by three.

Data: summary.json · SHA256 f1ad1ca99aea

Current status: unprinted and untested prototypes. Calculated release masses are candidates for measurement, not approved operating loads. The complete pressure/clearance transition and actual departure remain unverified.

The math behind the comparison

```
P = n m g,  g = 9.80665 m/s²
σ_axial = P/A;  σ_bending = M/Z
Z_rectangle = b t²/6;  Z_circle = π d³/32
P_section = chosen stress / stress per unit P
F_withdraw = c(μ, stroke, contact case) P + F_parasitic
m_release = min_stroke[(F_net − F_parasitic)/c] / (n g)
```

Mass means total carried-side mass: payload plus printed payload adapter plus all departing brackets and fasteners. Carrier-side coil/receiver/clamp mass is additional aircraft load. A 3g resultant reduces a given mass equivalent to one-third; it does not establish the correct aircraft load case.

What is assumed, and what is sourced

Input	Value / interpretation
Printed material screen	10 MPa tension/bending, 6 MPa shear, 10 MPa bearing; E=1500 MPa. Selected comparison values, not measured allowables.
Friction	μ=0.10-0.40 sensitivity; μ=0.20 main comparison. Actual printed surface friction is unmeasured.
Actuator reference	Adafruit 2776 lists 3 mm / 80 g at 5 V. Interpreted as gram-force: 0.080 × 9.80665 = 0.784532 N. No complete force curve or spring convention is supplied [S1].
Example force allocation	B=0.35 N. Hypothetical interpretation: minimum net pull 0.70 N with a 2:1 force ratio. Neither quantity is demonstrated; omitted drag reduces this allocation.
Material context	Bambu PETG HF typical tensile X-Y/Z: 34/23 MPa. Its 100% infill coupons were annealed/dried before testing; this does not establish the strength of the P6 prints [S2].
Boundary condition	Outward reaction through the defined retention axis, without extra payload-CG moment, side load, aircraft flexibility or aerodynamic torque.

The return spring must be counted once: subtract it when starting from gross magnetic force; do not subtract it again when using measured net external pull. A single catalog point is not a guaranteed minimum throughout the stroke.

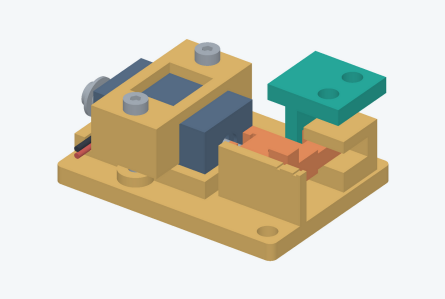
The new calculations include friction-induced yaw and, where relevant, pitch. Some older P6 force values omitted these moments and are superseded for comparison. The P6 CAD remains unchanged.

Straight keeper

The keeper is a 3 by 3 mm beam. Its overhung load creates opposed guide reactions; the pull is also offset from the loaded arm, so side-guide friction must be added.

Principal result
The overhang amplifies guide reactions.

Roof effective width 0.5-3 mm; contact insets



```
Z = 3 × 3² / 6 = 4.50 mm³
σ_keeper = P × 2.7 / 4.50 = 0.600 P MPa
R_near/P = 1 + 2.7/4 = 1.675; R_far/P = 0.675
D/P = μ(1 + 1.675 + 0.675) = 3.35μ
F/P = (D/P)[1 + 20μ/(11 − 7.5μ)]
At μ=0.20: F/P=0.952; inset contacts give 1.216
Roof-strip σ = (3.738 to 4.612)P / b_eff MPa
```

Selected structural calculations

Local mode	Stress coefficient	Screen threshold
proximal receiver roof strip bending	3.738 MPa/N	2.68 N
proximal receiver roof strip shear	1.142 MPa/N	5.25 N
keeper bending	0.600 MPa/N	16.67 N
keeper rectangular-section shear	0.167 MPa/N	36.00 N

This table uses b_eff=1 mm and retained nominal contacts. The first-page range also varies b_eff=0.5-3 mm, contact insets and N2 distal support effectiveness.

Release-mass example at μ = 0.20

```
m = 0.35 / (c × 9.80665)
c = 0.9521 to 1.2157 → 29.4 to 37.5 g at 1g
```

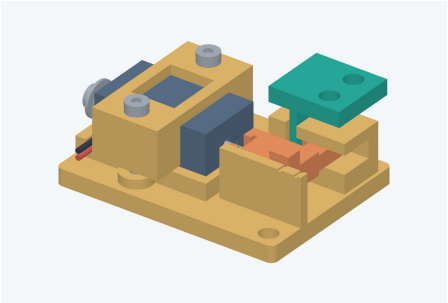
The roof load-spread width is unresolved. The isolated keeper bending result is not the assembly capacity. At the illustrative release budget, friction governs before these chosen retained-section thresholds.

Supported cross-bolt

A distal bridge shares load only while its 0.6 mm engagement remains. The governing pull calculation changes to an overhung keeper before the 1.4 mm lug clears.

Principal result
The second support is lost during withdrawal.

Roof effective width 0.5-3 mm; contact insets; distal support engaged/ineffective



```
At s=0.60 mm: a=1.0 mm; L=6.1 mm
R_near/P=1.16393; R_far/P=0.16393
D/P=2.32787μ; lateral span=11.6 mm
F/P=(D/P)[1 + 20μ/(11.6 − 7.5μ)]
At μ=0.20: peak F/P=0.650 nominal
No distal support + insets: F/P=0.784
Lug-floor Z=1.4 × 3.1²/6=2.242 mm³
σ_floor=0.775P/2.242=0.346P MPa
```

Selected structural calculations

Local mode	Stress coefficient	Screen threshold
proximal receiver roof strip bending	1.261 MPa/N	7.93 N
proximal receiver roof strip shear	0.385 MPa/N	15.57 N
distal roof spanning both Y walls	0.472 MPa/N	21.20 N
lug floor bending	0.346 MPa/N	28.93 N

This table uses b_eff=1 mm and retained nominal contacts. The first-page range also varies b_eff=0.5-3 mm, contact insets and N2 distal support effectiveness.

Release-mass example at μ = 0.20

```
m = 0.35 / (c × 9.80665)
c = 0.6500 to 0.7843 → 45.5 to 54.9 g at 1g
```

N2 improves the modeled pull force relative to N1, but its thinner lug and disappearing support prevent using the retained two-support condition as its release limit.

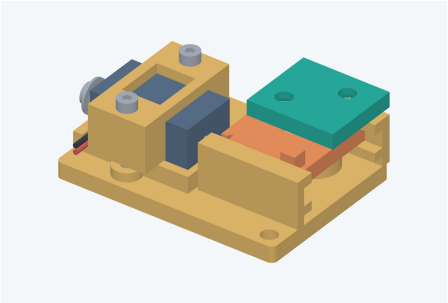
N3 / WORKED CALCULATION

Keyhole sliding gate

The actual head contact is a crescent, not a full annulus. CAD integration gives 7.551 mm² of retained contact and an eccentricity of 1.483 mm from the 3 mm neck axis.

Principal result
The neck bends because the head contact is eccentric.

Local gate-ear participation vs eccentric neck model



$A=\pi \times 3^2/4=7.069\ mm^2;\ Z=\pi \times 3^3/32=2.651\ mm^3$
$\sigma_{neck}=P/A + P \times 1.483/Z = 0.7011P\ MPa$
$P_{neck}=10/0.7011=14.26\ N$
$F/P=2\mu + 48\mu^2/(D - 14\mu)$
$D=17.4\ mm\ or\ 13.4\ mm\ for\ the\ contact\text{-}span\ sensitivity$
$At\ \mu=0.20:\ F/P=0.532\ to\ 0.581$

Selected structural calculations

Local mode	Stress coefficient	Screen threshold
Gate local ear, concentrated-load sensitivity	1.149 MPa/N	8.70 N
Neck, axial plus uniform-footprint eccentric bending	0.701 MPa/N	14.26 N
Gate, all load on uninterrupted right strip	0.649 MPa/N	15.40 N
Upper guide roof, inner-edge contact sensitivity	0.667 MPa/N	15.00 N

Axial neck tension alone suggests 70.69 N, but omits eccentric bending. The local gate-ear width and plate load transfer are unsolved; its 8.70 N value is a sensitivity, not a proved bound.

Release-mass example at μ = 0.20

$m = 0.35 / (c \times 9.80665)$
$c = 0.5315\ to\ 0.5811 \rightarrow 61.4\ to\ 67.1\ g\ at\ 1g$

A localized gate-ear model gives a lower 8.70 N threshold. The selected structural screens remain above the release-force examples. N3 is the simplest strong candidate for the next loaded-release test.

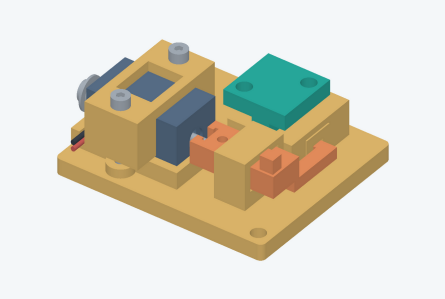
N4 / WORKED CALCULATION

Side-exit T-rail lock

The broad rail lips carry normal reaction P. The cross-stop carries the independent longitudinal reaction S. Pure normal gravity load supplies no force along the exit rail.

Principal result
Normal rail load and side-stop load are separate.

One rail vs equal rail sharing; separate stop load excluded



```
Each lip: b=7 mm, t=2.9 mm, cantilever a=3 mm
Z_lip=7 × 2.9²/6=9.812 mm³
σ_lip=(P/2) × 3/9.812=0.15288P MPa
P_normal=10/0.15288=65.41 N; one rail:32.71 N
Selected stop-wall screen: S≤6.525 N
Selected elastic sharing at μ=0.20: F≈0.786S
External slide drive Q must exceed rail drag, about μ_rail P
Travel along the rail must exceed 17.5 mm
```

Selected structural calculations

Local mode	Stress coefficient	Screen threshold
Rail lip root bending, equal sharing and edge contact	0.153 MPa/N P	65.41 N P
Adapter foot-wing bending, same 7 mm strip	0.143 MPa/N P	70.00 N P
Rail lip root maximum rectangular shear	0.037 MPa/N P	162.40 N P
Stop local guide wall cantilever	1.533 MPa/N S	6.52 N S

P is the normal rail reaction; S is the independent longitudinal stop reaction. The body-guide and local-guide simultaneous contact pattern is not established. The F/S example must not be converted to a unique payload rating.

If S/P=0.25, the selected F/S=0.7863 example gives P=0.35/(0.7863×0.25)=1.78 N, or 182 g normal-equivalent at 1g. If S/P=1, it gives 45 g. These demonstrate the dependence on the extractor; they are not normal-drop capacities.

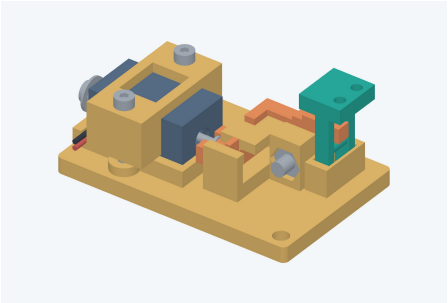
N4 is the strongest normal-holding candidate in these local section models. Its release coefficient is an assumed elastic load-sharing scenario, not a proven simultaneous rigid contact state. It has no unique normal-drop mass rating.

Pivoting hook + short sear

The load and heel act on opposite sides of the pivot. The nominal heel reaction is 6/8.7=0.690 times P. Guide roll and yaw add to the sear drag.

Principal result
The sear is loaded, and the hook arm sees torsion.

Selected linear bending/torsion interaction; alternate interactions are in the full record



```
Heel arm: b=3.0 mm; t=2.7 mm; Z=3.645 mm³
σ_b=1.22984P MPa; τ_t=0.573292P MPa
P(1.22984/10 + 0.573292/6)≤1
Selected linear interaction: P≤4.576 N
Heel ratio q=6/(36−x_contact)
D_yaw=L_side−3.5μ; H=μ(q y_h + A y_A + B y_B)/D_yaw
F/P=μ(q+A+B+2H)
At μ=0.20: full-stroke F/P=0.507 to 0.710
```

Selected structural calculations

Local mode	Stress coefficient	Screen threshold
Hook heel arm bending + torsion, linear interaction	1.230 bending + 0.573 shear	4.58 N
Integral printed cross-pin bending, conservative point load at Y10	1.123 MPa/N	8.90 N
Hook load shelf bending	0.844 MPa/N	11.84 N
Heel lateral projection bending	0.502 MPa/N	19.94 N

The linear bending/torsion interaction is a selected material screen, not a validated orthotropic failure criterion. The alternative quadratic screen is 6.42 N. The sear reduction omits the beneficial heel-friction closing moment.

Release-mass example at μ = 0.20

```
m = 0.35 / (c × 9.80665)
c = 0.5075 to 0.7101 → 50.3 to 70.3 g at 1g
```

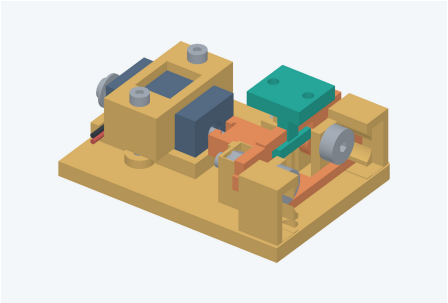
N5 has the lowest nominal pull coefficient, but its contact sensitivity overlaps N3 and its thin hook arm has a lower structural screen. Opening still needs an external moment after the sear clears.

Paired jaws + common sear

Each jaw has a small cantilever shelf. Equal sharing doubles the ideal shelf capacity; one-jaw loading removes that benefit and adds yaw to the sear. The pull line below the fingers also creates pitch.

Principal result
Two jaws add sensitivity to uneven loading.

One-jaw vs equal-jaw shelf loading



```
Jaw shelf: b=2.2 mm; t=2.3 mm; a=3.1 mm
Z=2.2 × 2.3²/6=1.940 mm³
σ_jaw=P_i × 3.1/1.940=1.5982P_i MPa
P_i≤10/1.5982=6.257 N
Equal sharing: P≤12.514 N; one jaw: P≤6.257 N
Total heel ratio q=5.2/5.35=0.97196
Equal-share pitch: B/P=q(a+6μ)/(L−6μ)
F/P=2μ(q+B/P), before asymmetric yaw
At μ=0.20: selected F/P=0.715 to 1.702
```

Selected structural calculations

Local mode	Stress coefficient	Screen threshold
Inner jaw shelf bending	1.598 MPa/N	6.26 N
Adapter stem tension plus one-sided head eccentricity	0.974 MPa/N	10.26 N
Adapter head shoulder bending	0.620 MPa/N	16.13 N
Jaw heel cantilever bending	0.957 MPa/N	10.45 N

This structural table places all normal load on one jaw. Equal sharing doubles the ideal shelf limit. Metal shoulder screw stress is below the modeled printed-section concern; actual mounts remain outside the calculation.

Release-mass example at μ = 0.20

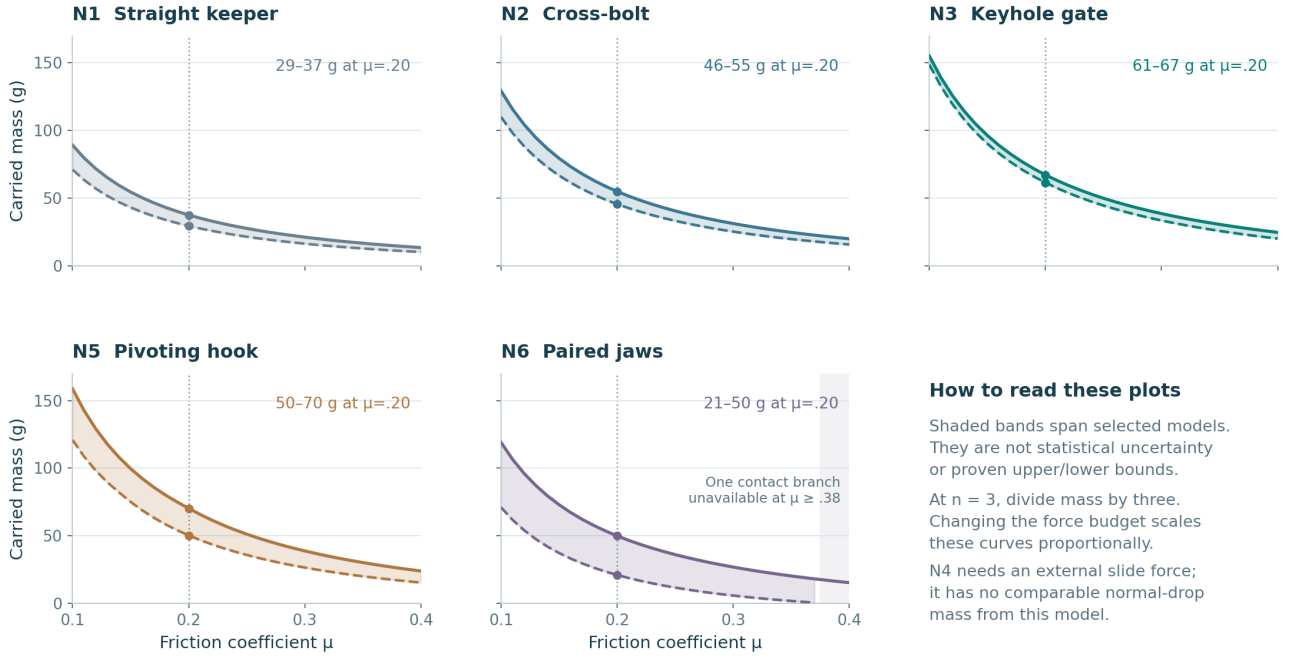
```
m = 0.35 / (c × 9.80665)
c = 0.7149 to 1.7021 → 21.0 to 49.9 g at 1g
```

More printed pieces do not buy better release performance here. Both jaws must move, and some high-friction asymmetric contact branches have no valid sliding solution.

Friction can dominate the choice

Friction changes the answer sharply

Conditional release mass vs friction coefficient · budget 0.35 N · n = 1



Neither net coil pull nor printed-part friction has been measured. Opening and departure still require separate verification.

Mass includes payload, adapter and payload-side hardware. Geometry: P6-01 unchanged; analytical study: P6-C1.

Data: summary.json · SHA256 f1ad1ca99aea

The plotted bands span selected contact cases only. They do not bound all real printed mechanisms. N6 has an unresolved high-friction asymmetric branch; missing values must not be interpreted as zero force or zero load.

Change	Effect in the stated model
Double allocated net force	Doubles the friction-limited mass until another structural or departure limit becomes active.
1g to 3g resultant	Divides the corresponding carried mass by three.
Increase friction	Reduces release mass, often faster than 1/μ because guide side/pitch reactions feed back into friction.
Change printed strength	Changes the section screen, but does not reduce the friction coefficient or increase coil force.
Move payload CG off-axis	Adds bending and extra guide reactions. The centered-load table no longer describes that installation.
External force applied after unlock	Can unload N4's stop during withdrawal, but requires an extractor and sequencing that P6 does not include.

Design implication: improving a tiny printed section does not fix a loaded keeper that the coil cannot withdraw. A higher-force actuator or a latch geometry that reduces the actuator load is the relevant next design change if substantially larger release loads are required.

What has actually been established

Evidence	Result and limit
N1/N2 geometry	Issued STEP checks passed for critical sections and identities. The independent force/moment reconstruction passed 23,456 sampled states.
N3/N4 geometry and review	Ten issued-STEP checks passed. Independent review accepted the retained arithmetic and flagged the N4 simultaneous-contact issue; its reporting scope was corrected.
N5/N6 geometry and balance	Nine critical sections and 280 candidate opposing patches passed. Sear force/moment residuals were below 3.5×10^{-13} in the tested branches. Candidate patches do not prove simultaneous loaded contact.
FEA	No new structural FEA was performed here. Prior P5 FEA covers only the N1-equivalent isolated adapter and cannot rate a full mechanism.
Physical performance	No parts, net coil force, loaded releases, fatigue, creep, heat exposure or aircraft installations were physically tested.

Measurements that turn this into a capacity decision

1. Measure minimum net external coil pull versus stroke at the actual supply voltage and warm condition; establish the return-spring convention.

2. Measure unloaded and loaded withdrawal force through the full stroke for each printed candidate, starting below the predicted screen and recording friction/fit variation.

3. Verify that the adapter clears before power ends. For N5/N6 observe actual opening, and for N4 provide and measure sideward extraction.

4. Inspect the last-overlap edge for indentation, catching and permanent set. A shrinking ideal contact patch gives an unbounded constant-load pressure calculation.

5. Qualify the selected material, mounting stack, payload CG, flight load factors, wear and thermal/creep exposure before assigning a working load.

Primary sources and reproducibility

[S1] Adafruit 2776 product technical details
https://www.adafruit.com/product/2776

[S2] Bambu Lab PETG HF Technical Data Sheet v1.0
https://store.bblcdn.com/3a230e260a3a47c2b0db0156e07eef91.pdf

[S3] P6 issued CAD and audit package
https://compact-modular-hardpoint-r1.drunken.chatgpt.site

The companion ZIP includes the three independent calculation scripts, detailed derivations, scalar results, CSV tables, geometry checks, fixed STEP inputs, the aggregation code, plot source and reproduction instructions. Read METHOD.md and REPRODUCE.md before altering assumptions.